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Questions

Q1.

This question concerns acidic solutions.

(a) (i) A solution of hydrochloric acid has a pH of 1.125.

Calculate the concentration of hydrogen ions and of hydroxide ions, in mol dm^{-3}, in this solution.
[$K_w = 1.00 \times 10^{-14}$ mol^2 dm^{-6}]

(2)

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(ii) Calculate the volume of water which must be added to 25.0 cm^3 of this solution to increase the pH from 1.125 to 1.500, giving your answer to an appropriate number of significant figures.

(4)

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(b) Phosphoric(V) acid, H_3PO_4, is a weak acid, which dissolves in water, giving the equilibrium

Equation 1 $H_3PO_4(aq) + H_2O(l) \rightleftharpoons H_2PO_4^-(aq) + H_3O^+(aq)$

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The $H_2PO_4^-$ ion formed when phosphoric(V) acid is added to water can dissociate further into HPO_4^{2-}.

Equation 2 $H_2PO_4^-(aq) + H_2O(l) \rightleftharpoons HPO_4^{2-}(aq) + H_3O^+(aq)$

(i) Identify the acids and bases in the spaces below Equation 2, linking the acid-base conjugate pairs.

(2)

(ii) Explain why very little dissociation of the $H_2PO_4^-$ ion occurs in solutions of phosphoric(V) acid.

(2)

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(c) (i) Using Equation 1, write an expression for the acid dissociation constant, K_{a1}, for phosphoric(V) acid, H_3PO_4.

(1)

(ii) A 0.500 mol dm^{-3} solution of phosphoric(V) acid has a pH of 1.20.
Calculate the value of K_{a1}, stating its units.
Assume that there is no further dissociation of the $H_2PO_4^-$ ion.

(4)

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(d) Phosphate buffer solutions containing hydrogenphosphate ions, HPO_4^{2-}, and dihydrogenphosphate ions, $H_2PO_4^-$, are commonly used in biological research.

Explain, using a relevant ionic equation, how a solution containing these ions can act as a buffer when a small amount of **alkali** is added.

(3)

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(Total for question = 18 marks)

Q2.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A buffer solution contains ethanoic acid, with a concentration of 0.10 mol dm^{-3}, and sodium ethanoate, with a concentration of 0.050 mol dm^{-3}.

K_a for ethanoic acid = 1.7×10^{-5} mol dm^{-3}

The pH of this buffer solution is

- ☐ **A** 2.88
- ☐ **B** 4.47
- ☐ **C** 4.77
- ☐ **D** 5.07

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(Total for question = 1 mark)

Q3.

A solution of sodium hydroxide of concentration 0.0080 mol dm^{-3} has a pH

- **A** between 7 and 9
- **B** between 9 and 11
- **C** between 11 and 13
- **D** above 13

(Total for question = 1 mark)

Q4.

How are 20 cm^3 of 0.05 mol dm^{-3} $H_2SO_4(aq)$ and 20 cm^3 of 0.10 mol dm^{-3} $CH_3COOH(aq)$ alike?

Both solutions

- **A** have the same pH
- **B** have the same total concentration of ions
- **C** have the same total concentration of **negative** ions
- **D** require 20 cm^3 of 0.10 mol dm^{-3} NaOH(aq) for complete reaction

(Total for question = 1 mark)

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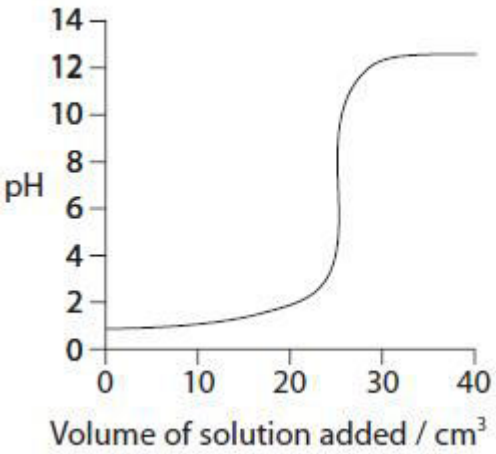
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Q5.

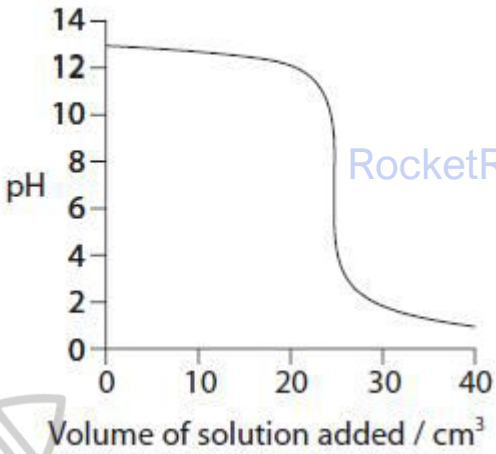
Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The graphs show the changes in pH during a series of titrations.
Each titration involves two solutions, each of concentration 0.1 mol dm^{-3}.

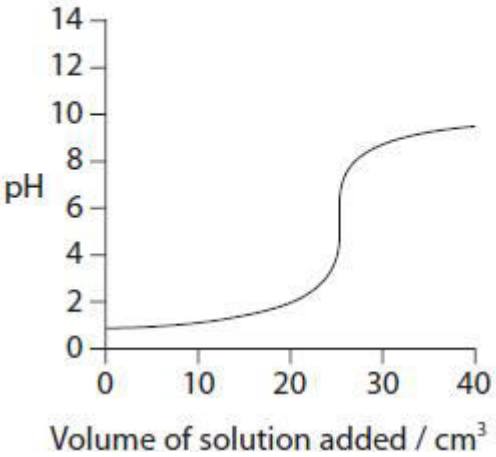
Graph **1**



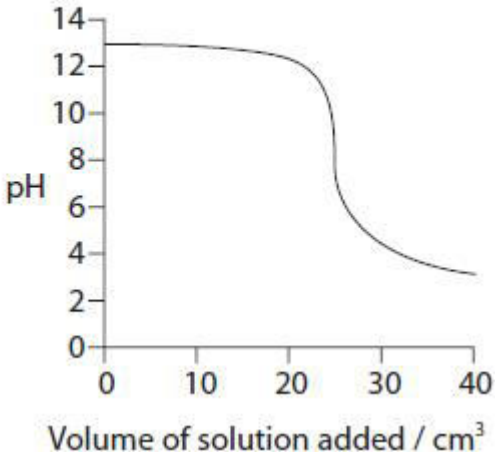
Graph **2**



Graph **3**



Graph **4**



(a) Which graph has ethanoic acid as one of the reactants?

(1)

- ☒ **A** Graph **1**
- ☒ **B** Graph **2**
- ☒ **C** Graph **3**
- ☒ **D** Graph **4**

(b) Which graph shows a solution that has an acidic pH at the end-point?

(1)

- ☒ **A** Graph **1**
- ☒ **B** Graph **2**
- ☒ **C** Graph **3**
- ☒ **D** Graph **4**

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(c) Which indicator is suitable for the titration shown in Graph **3**?

Use your Data Booklet.

(1)

- ☒ **A** bromocresol green
- ☒ **B** bromothymol blue
- ☒ **C** phenol red
- ☒ **D** thymol blue (acid)

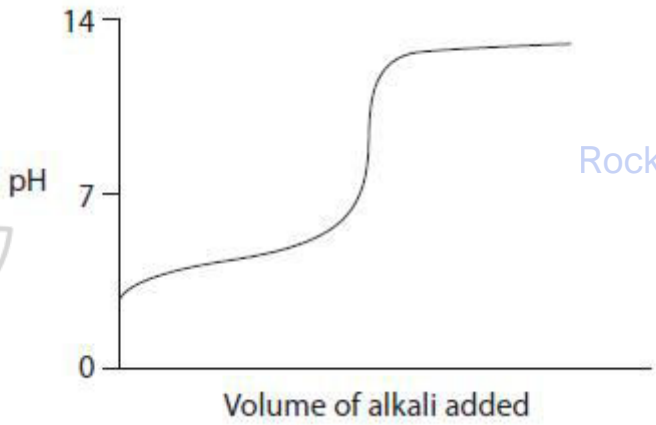
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(Total for question = 3 marks)

Q6.

Which two solutions, both of concentration 0.10 mol dm^{-3}, have been used to produce this titration curve?



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- ☒ **A** HCl(aq) and KOH(aq)
- ☒ **B** HCOOH(aq) and KOH(aq)
- ☒ **C** HCl(aq) and NH_3(aq)
- ☒ **D** HCOOH(aq) and NH_3(aq)

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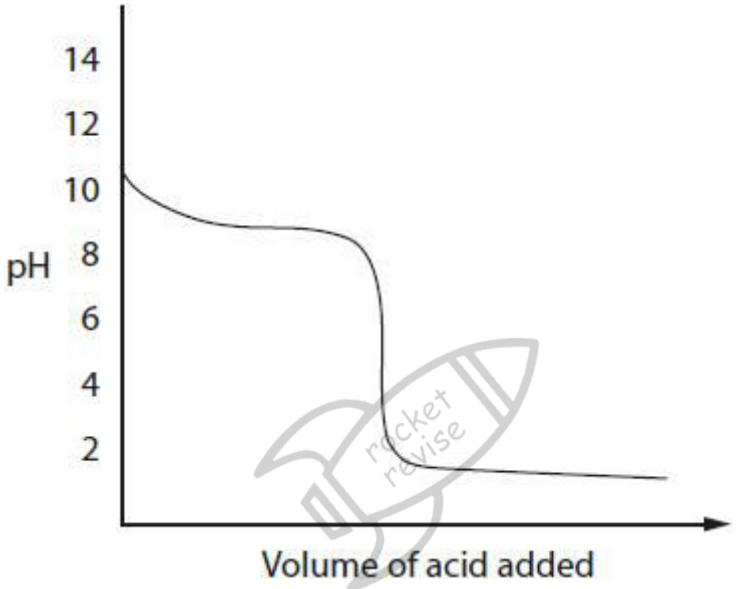
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(Total for question = 1 mark)

Q7.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

A graph of pH against volume of acid added for an acid-base titration is shown.



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(a) Which acidic solution was used in the titration?

(1)

- ☐ **A** 0.1 mol dm^{-3} CH_3COOH
- ☐ **B** 1.0 mol dm^{-3} CH_3COOH
- ☐ **C** 0.1 mol dm^{-3} HCl
- ☐ **D** 1.0 mol dm^{-3} HCl

(b) Which basic solution was used in the titration?

(1)

- ☐ **A** NH_3
- ☐ **B** LiOH
- ☐ **C** $Ba(OH)_2$
- ☐ **D** NaOH

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(c) A student suggested five indicators that might be used in this titration:

thymol blue
methyl orange
bromophenol blue
bromocresol green
phenolphthalein

How many of these indicators would be suitable? Use your Data Booklet.

(1)

- ☐ **A** 5
- ☐ **B** 4
- ☐ **C** 3

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☒ **D** 2

(Total for question = 3 marks)

Q8.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

When ethanoic acid and chloroethanoic acid are mixed, an equilibrium is set up.

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$$CH_3COOH + CH_2ClCOOH \rightleftharpoons CH_3COOH_2^+ + CH_2ClCOO^-$$

The Brønsted-Lowry acids in this equilibrium are

☒ **A** CH_3COOH and $CH_2ClCOOH$

☒ **B** CH_3COOH and $CH_3COOH_2^+$

☒ **C** $CH_2ClCOOH$ and $CH_3COOH_2^+$

☒ **D** $CH_3COOH_2^+$ and CH_2ClCOO^-

(Total for question = 1 mark)

Q9.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

A diprotic acid, H_2A, was titrated with sodium hydroxide solution.

$$H_2A(aq) + 2NaOH(aq) \rightarrow Na_2A(aq) + 2H_2O(l)$$

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A 25.0 cm^3 portion of 0.100 mol dm^{-3} sodium hydroxide solution required 12.80 cm^3 of the solution of the diprotic acid for complete neutralisation.

What is the concentration of H_2A in mol dm^{-3}?

☒ **A** 2.56×10^{-2}

☒ **B** 9.77×10^{-2}

☒ **C** 1.95×10^{-1}

☒ **D** 3.91×10^{-1}

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(Total for question = 1 mark)

Q10.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the pH of 0.010 mol dm^{-3} aqueous calcium hydroxide, $Ca(OH)_2(aq)$?

[pK_w = 14]

- ☒ **A** 11.7
- ☒ **B** 12.0
- ☒ **C** 12.3
- ☒ **D** 13.3

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(Total for question = 1 mark)

Q11.

This question is about sulfuric acid and its salts.

(a) The manufacture of sulfuric acid involves the equilibrium

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ $\Delta_r H = -197$ kJ mol^{-1}

(i) A catalyst of vanadium(V) oxide is used in this reaction.
State the effect, if any, of the catalyst on the value of the equilibrium constant, K_p .

(1)

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(ii) The temperature used for this reaction in industry is 700 K.
Explain, in terms of the equilibrium constant and the equilibrium position, the effect of an increase in temperature on the equilibrium yield of sulfur trioxide.

(2)

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(iii) Write the expression for the equilibrium constant, K_p , for this equilibrium. State symbols are not required.

(1)

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(iv) A mixture of 2.00 mol of sulfur dioxide and 1.00 mol of oxygen is allowed to reach equilibrium at 5.00 atm pressure.
1.60 mol of sulfur trioxide is formed.
Calculate the value of K_p.
Include units and give your answer to an appropriate number of significant figures.

(4)

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(b) Sulfur trioxide is used to produce sulfuric acid.

(i) Commercial concentrated sulfuric acid contains 98.5 % H_2SO_4 and 1.5 % water by mass.
The density of concentrated sulfuric acid is 1800 g dm^{-3}.
Calculate the concentration of this sulfuric acid in mol dm^{-3}.

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(2)

(ii) The pH of a 0.10 mol dm^{-3} solution of sulfuric acid at 25 °C is 0.97.
Calculate the concentration of hydrogen ions, in mol dm^{-3}, in this solution.

(1)

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(iii) In an aqueous solution of sulfuric acid, the following equilibria exist.

$H_2SO_4(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + HSO_4^-(aq)$ K_a very large

$HSO_4^-(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + SO_4^{2-}(aq)$ $K_a = 0.012\ mol\ dm^{-3}$

Explain, in terms of these equilibria, why the concentration of hydrogen ions in a 0.10 mol dm^{-3} solution of sulfuric acid is **not** 0.20 mol dm^{-3}. No calculation is required.

(2)

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(c) A buffer solution is made from HSO_4^- and SO_4^{2-} ions.

(i) Write two ionic equations involving HSO_4^- and SO_4^{2-} ions to show how this solution acts as a buffer.
State symbols are not required.

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(2)

(ii) A buffer solution is formed by mixing
25.0 cm^3 of a solution that is 0.150 mol dm^{-3} with respect to SO_4^{2-} ions with
75.0 cm^3 of a solution that is 0.100 mol dm^{-3} with respect to HSO_4^- ions.
Calculate the pH of this buffer solution.
[K_a for HSO_4^- ions = 0.012 mol dm^{-3}]

(5)

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(Total for question = 20 marks)

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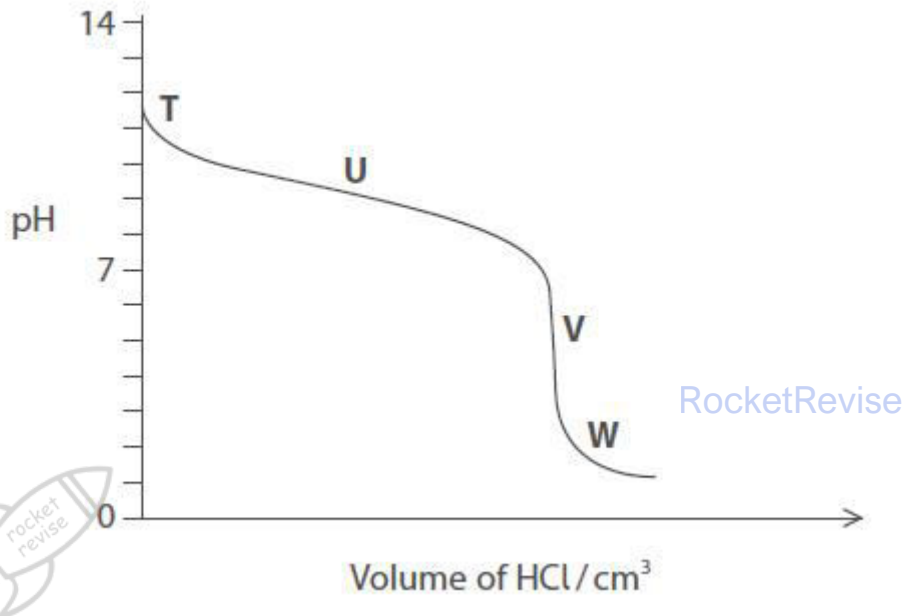
Q12.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

A titration was carried out by adding 0.1 mol dm^{-3} hydrochloric acid to 0.1 mol dm^{-3} aqueous ammonia.

$$HCl(aq) + NH_3(aq) \rightarrow NH_4Cl(aq)$$

The titration curve is shown.



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(a) Which region of the graph represents the most effective buffer solution?

(1)

- ☒ **A** region **T**
- ☒ **B** region **U**
- ☒ **C** region **V**
- ☒ **D** region **W**

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(b) Which of these is the best indicator to use in this titration?

[Refer to the Data Booklet]

(1)

- ☒ **A** methyl red
- ☒ **B** phenol red
- ☒ **C** phenolphthalein
- ☒ **D** thymol blue

(c) What is the approximate pH of an ammonium chloride solution?

(1)

- ☒ **A** 2.0
- ☒ **B** 5.8
- ☒ **C** 9.7
- ☒ **D** 11.3

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(Total for question = 3 marks)

Q13.

Sodium hydrogensulfate is a widely used acid, with applications that include removing limescale and as a food additive. Sodium hydrogensulfate is a weak acid because of the presence of the hydrogensulfate ion, HSO_4^-.

(a)(i) Write the equation for the dissociation of the hydrogensulfate ion in aqueous solution. State symbols are not required.

(1)

(ii) A solution of sodium hydrogensulfate has pH = 1.13
Calculate the concentration of this solution, in g dm^{-3}.
[pK_a of HSO_4^- = 1.92]

(5)

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(iii) State the assumptions you have used in (a)(ii).

(2)

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(b) A solution containing sodium hydrogensulfate and sodium sulfate is a buffer that is used to preserve urine for steroid analysis.

(i) State what is meant by the term buffer.

(2)

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(ii) Calculate the pH of the buffer prepared by dissolving 0.750 mol of sodium hydrogensulfate and 0.500 mol of sodium sulfate in distilled water to make 1.00 dm^3 of solution.

(3)

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(iii) Separate samples of 0.00500 mol of hydrochloric acid are added to 1.00 dm^3 of distilled water and to the buffer in (b)(ii).

Calculate the pH **changes** that result in each case.

Assume that the volumes remain constant at 1.00 dm^3.

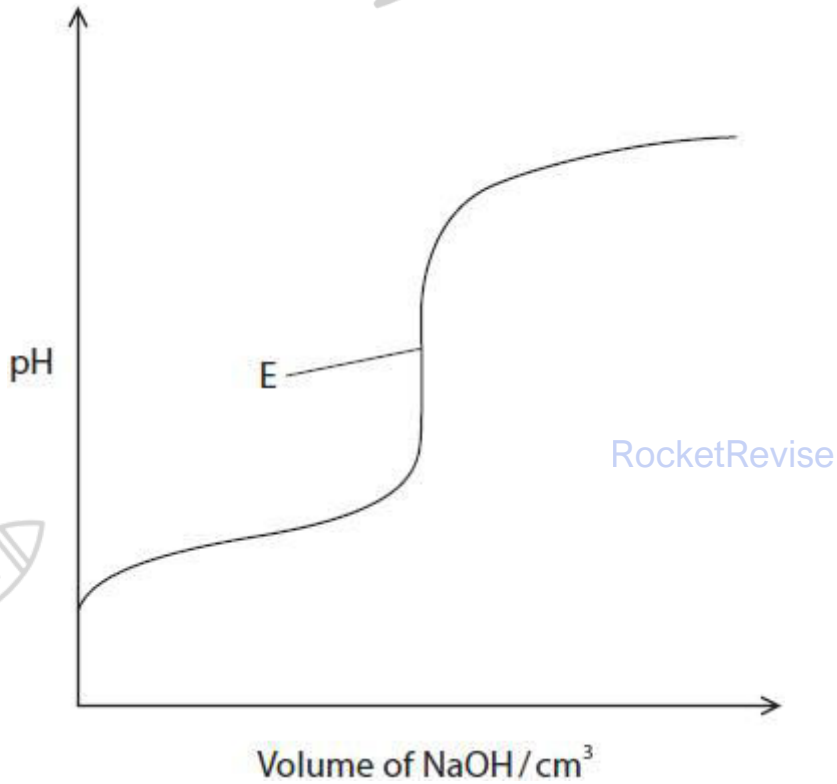
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(c) The titration curve obtained from the addition of sodium hydroxide solution to a weak acid is shown. The equivalence point (E) of this titration occurred at pH = 8



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Explain the observations that would be made if methyl orange (pK_{In} = 3.7) were used as the indicator for this titration.

(3)

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(Total for question = 20 marks)

Q14.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

This question is about weak acids.

pK_a of ethanoic acid, CH_3COOH = 4.8

pK_a of chloroethanoic acid, $CH_2ClCOOH$ = 2.9

(a) What is the pH of a 0.100 mol dm^{-3} solution of chloroethanoic acid?

(1)

- ☐ **A** 0.27
- ☐ **B** 1.95
- ☐ **C** 2.90
- ☐ **D** 3.90

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(b) Which is the acid-conjugate base pair in the reaction between ethanoic acid and chloroethanoic acid?

(1)

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	Acid	Conjugate base
☐ A	CH_3COOH	CH_3COO^-
☐ B	CH_3COOH	$CH_3COOH_2^+$
☐ C	$CH_2ClCOOH$	CH_2ClCOO^-
☐ D	$CH_2ClCOOH$	$CH_2ClCOOH_2^+$

(Total for question = 2 marks)

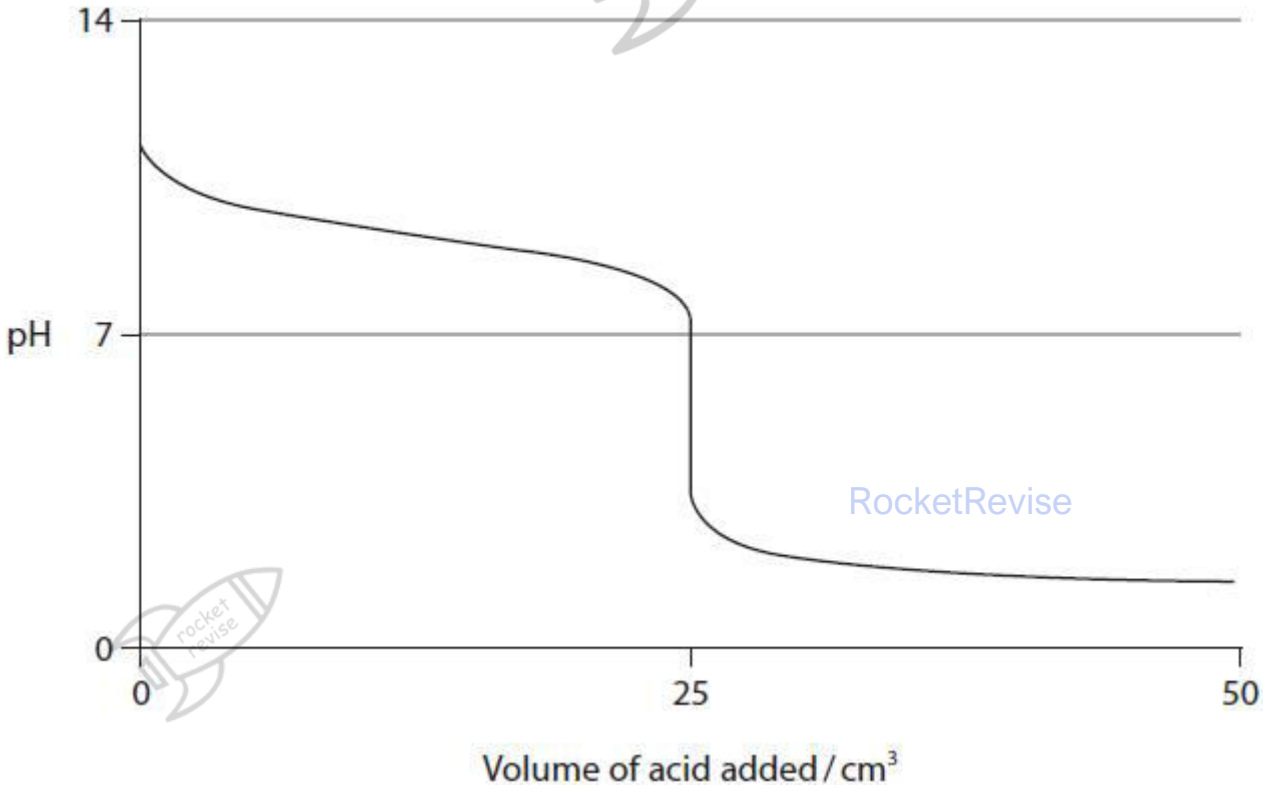
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Q15.

Answer the questions with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A 0.1 mol dm^{-3} aqueous acid solution was added to 25 cm^3 of a 0.1 mol dm^{-3} aqueous solution of an alkali.

The titration curve shown was obtained.



(a) Which titration would produce this curve?

(1)

☐ **A** ethanoic acid added to sodium hydroxide
☐ **B** hydrochloric acid added to sodium hydroxide
☐ **C** ethanoic acid added to ammonia

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☒ **D** hydrochloric acid added to ammonia

(b) Which indicator would be **most** suitable for this titration?

(1)

	Indicator	pH range
☒ **A**	malachite green	0.2–1.8
☒ **B**	methyl yellow	2.9–4.0
☒ **C**	methyl red	4.2–6.3
☒ **D**	thymol blue	8.0–9.6

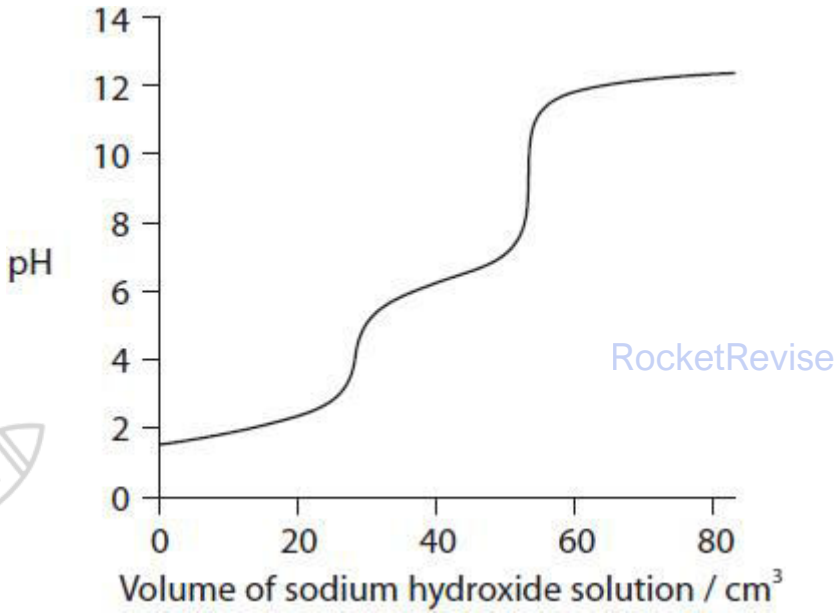
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(Total for question = 2 marks)

Q16.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

An example of a diprotic acid is *cis*-butenedioic acid. Titration of this acid using sodium hydroxide solution gave the titration curve shown.



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Which indicator would be most suitable for measuring the end-point of the neutralisation of the proton in *cis*-butenedioic acid which has a pK_a = 6.33?

Use your Data Booklet.

☒ **A** bromocresol green
☒ **B** bromothymol blue

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☒ **C** litmus
☒ **D** phenolphthalein

(Total for question = 1 mark)

Q17.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the pH of a solution containing 0.100 mol dm^{-3} CH_3COOH and 0.200 mol dm^{-3} CH_3COONa ?

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[pK_a CH_3COOH = 4.76]

☒ **A** 5.06
☒ **B** 4.76
☒ **C** 4.46
☒ **D** 2.88

(Total for question = 1 mark)

Q18.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the pH of a 0.050 mol dm^{-3} solution of barium hydroxide, $Ba(OH)_2$?

☒ **A** 13.3
☒ **B** 13.0
☒ **C** 12.7
☒ **D** 12.4

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(Total for question = 1 mark)

Q19.

The equation for the formation of ammonia in the Haber Process is shown

$$\tfrac{1}{2}N_2(g) + 1\tfrac{1}{2}H_2(g) \rightleftharpoons NH_3(g)$$

(a) At 298 K the standard entropy change of the system, $\Delta S^{\ominus}_{system} = -98\,J\,K^{-1}\,mol^{-1}$.

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Calculate the standard entropy of one mole of ammonia.

Use the value of $\Delta S^{\ominus}_{system}$ and the data in the table.

Substance	Standard molar entropy, $S^{\ominus}$ / $JK^{-1}mol^{-1}$
N_2	192
H_2	131

(2)

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(b) The value of the total entropy change, ΔS_{total}, varies with temperature.

Data for the value of ΔS_{total} at different temperatures but at standard pressure of 100 kPa are given for this reaction.

Temperature / K	$1/T$ / K^{-1}	ΔS_{total} / $kJ\,K^{-1}mol^{-1}$
250	4.00×10^{-3}	8.27×10^{-2}
375	2.67×10^{-3}	2.25×10^{-2}
500	2.00×10^{-3}	-0.764×10^{-2}
625	1.60×10^{-3}	-2.57×10^{-2}
750	1.33×10^{-3}	-3.77×10^{-2}

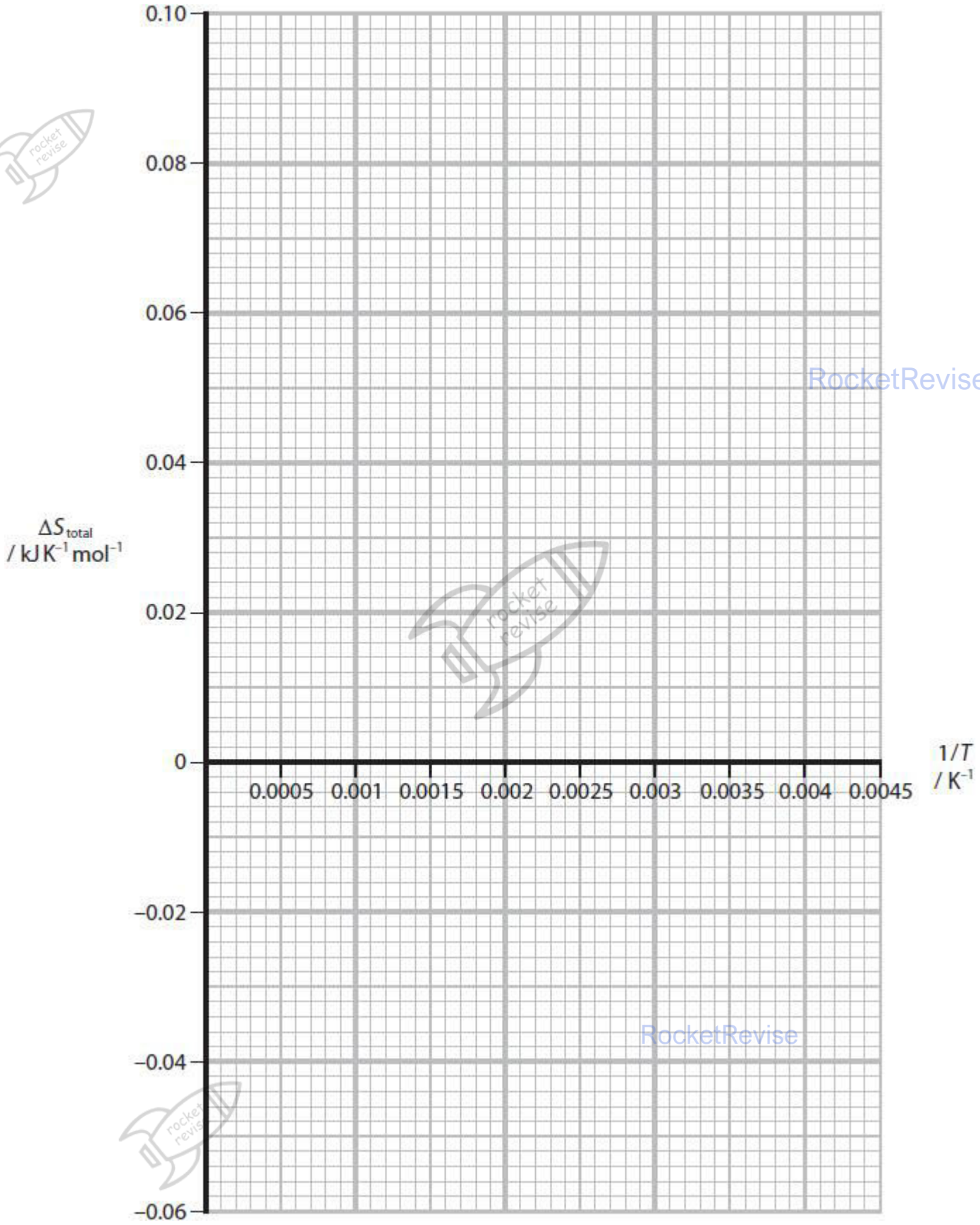
Plot a graph of ΔS_{total} against $1/T$ on the grid.
Include a line of best fit.

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(2)

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(c) The relationship between ΔS_{total} and $1/T$ can be found by combining the two equations:

$\Delta S_{total} = \Delta S_{surroundings} + \Delta S_{system}$

and $\Delta S_{surroundings} = -\Delta H/T$

to give

$\Delta S_{total} = -\Delta H/T + \Delta S_{system}$

(i) Determine the gradient of the line plotted in (b), including units in your answer.

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(1)

(ii) Identify the thermodynamic quantity that can be obtained from this gradient.

(1)

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(iii) Determine the temperature at which the reaction ceases to be thermodynamically feasible at a pressure of 100 kPa.

(1)

.....

(d) The industrial synthesis of ammonia

$$\tfrac{1}{2}N_2(g) + 1\tfrac{1}{2}H_2(g) \rightleftharpoons NH_3(g)$$

is carried out at pressures of about 20 000 kPa and temperatures between 700K and 750K. These temperatures are higher than the answer to (c)(iii).

(i) State the relationship between the total entropy, ΔS_{total}, and the equilibrium constant, K.

(1)

.....

(ii) Calculate the value of the equilibrium constant K at 750K.
[ΔS_{total} at 750K = –37.7 J K^{-1} mol^{-1}]

(2)

(iii) Explain why ΔS_{total} decreases with an increase in temperature.

(3)

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(iv) State how the Haber Process is made economically feasible at 750 K even though the total entropy change is negative.

(1)

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(e) Ammonia from the Haber Process reacts with acids.

With phosphoric acid, H_3PO_4, a number of products are formed in solution.
One of these is the fertiliser diammonium hydrogenphosphate.

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(i) Write an equation for the production of this fertiliser.
State symbols are not required.

(2)

(ii) Write an **ionic** equation to show that ammonium ions are acidic in aqueous solution.
State symbols are not required.

(1)

(iii) A solution containing both ammonia and ammonium ions acts as a buffer.
Explain, using a relevant ionic equation, the effect of adding a small amount of acid to this buffer.

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(3)

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(Total for question = 20 marks)

Q20.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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Which of these indicators is **most** suitable for the titration of ammonia solution with hydrochloric acid?

- ☐ **A** methyl violet ($pK_{in} = 0.8$)
- ☐ **B** bromocresol green ($pK_{in} = 4.7$)
- ☐ **C** thymol blue ($pK_{in} = 8.9$)
- ☐ **D** alizarin yellow ($pK_{in} = 12.5$)

(Total for question = 1 mark)

Q21.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Propanone reacts with iodine in the presence of a sulfuric acid catalyst.

$$CH_3COCH_3(aq) + I_2(aq) \rightarrow CH_3COCH_2I(aq) + HI(aq)$$

(a) Which method would be most suitable for following this reaction?

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(1)

- ☐ **A** colorimetry
- ☐ **B** timing how long the solution takes to go cloudy
- ☐ **C** quenching and titrating with an acid
- ☐ **D** quenching and titrating with starch solution

(b) The rate equation for this reaction is:

$$\text{rate} = k[H^+][CH_3COCH_3]$$

Two experiments were carried out. In both experiments the concentration of propanone and iodine remained the same but the concentration of the sulfuric acid was changed.

The initial rate in the first experiment was three times faster than the initial rate in the second experiment.

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In the first experiment the initial pH was 1.50

What is the initial pH in the second experiment?

(1)

- ☒ **A** 1.02
- ☒ **B** 1.98
- ☒ **C** 2.28
- ☒ **D** 4.50

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(Total for question = 2 marks)

Q22.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

At 25 °C, the pH of pure water is 7.00 and at 100 °C, the pH of pure water is 6.14. What can be deduced from this information?

	Enthalpy change of dissociation of water	Concentration of hydrogen ions
☐ **A**	endothermic	higher at 100 °C than at 25 °C
☐ **B**	endothermic	lower at 100 °C than at 25 °C
☐ **C**	exothermic	higher at 100 °C than at 25 °C
☐ **D**	exothermic	lower at 100 °C than at 25 °C

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(Total for question = 1 mark)

Q23.

Answer the question with a cross in the box you think is correct ☒. If you change your mind

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about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the pH of a 0.27 mol dm^{-3} aqueous solution of sodium hydroxide?

[K_w = 1.0 × 10^{-14} mol2 dm^{-6} at 298 K]

- ☐ **A** 0.57
- ☐ **B** 13.43
- ☐ **C** 13.73
- ☐ **D** 14.57

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(Total for question = 1 mark)

Q24.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the pH of a 0.010 mol dm^{-3} aqueous solution of carbonic acid?

[K_a of carbonic acid = 4.5 × 10^{-7} mol dm^{-3} at 298 K]

- ☐ **A** 2.00
- ☐ **B** 4.17
- ☐ **C** 6.35
- ☐ **D** 8.35

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(Total for question = 1 mark)

Q25.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

In which equation is H_2O acting **only** as a Brønsted-Lowry acid?

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☒ **A** $CH_3COOH + H_2O \rightarrow CH_3COO^- + H_3O^+$

☒ **B** $NH_4^+ + H_2O \rightarrow H_3O^+ + NH_3$

☒ **C** $NH_3 + H_2O \rightarrow NH_4^+ + OH^-$

☒ **D** $2H_2O \rightarrow H_3O^+ + OH^-$

(Total for question = 1 mark)

Q26.

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This question is about ammonia and ammonium chloride.

(a) Ammonia is produced by reacting nitrogen and hydrogen as shown.

$$N_2(g) + 3H_2(g) \leftrightarrow 2NH_3(g)$$

(i) Write the expression for the equilibrium constant, K_p.

(1)

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(ii) An equilibrium was established by mixing nitrogen and hydrogen in a 1 : 3 ratio by volume, at a temperature of 450 °C and pressure of 200 atm.
The equilibrium mixture contained 28 % of ammonia by volume.
Complete the table.

(3)

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Substance	Mole fraction	Partial pressure / atm
N_2		
H_2		
NH_3	0.28	56

(iii) Calculate the equilibrium constant.
Give your answer to an appropriate number of significant figures and include units, if required.

(3)

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(iv) When the temperature was reduced to 400 °C at the same pressure, the equilibrium mixture contained 36 % of ammonia.
Explain what can be deduced about this reaction from this information.

(2)

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(b) Ammonium chloride can be produced by reacting ammonia with hydrogen chloride.

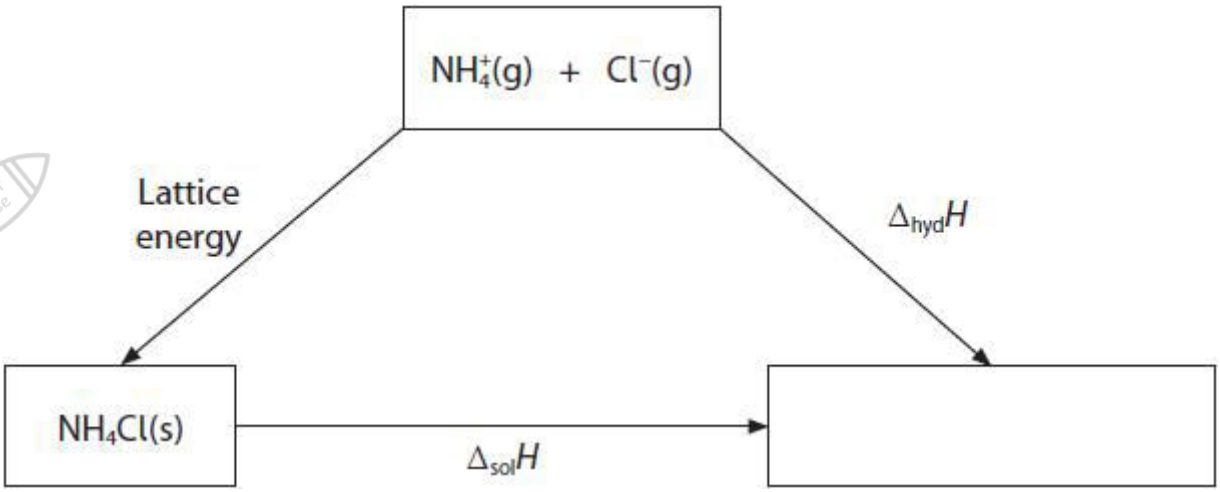
Ammonium chloride is a white solid that is very soluble in water.
Hess's Law can be used to calculate the enthalpy of solution of ammonium chloride, using hydration enthalpies and lattice energy.

(i) Complete the Hess cycle by filling in the empty box.

(1)

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(ii) Complete the expression for the enthalpy change of solution using the hydration enthalpies and lattice energy.

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(1)

$\Delta_{sol}H$ =

(iii) Calculate the enthalpy change of solution, using your expression in (b)(ii) and these data.

Enthalpy change of hydration of NH_4^+ = −307 kJ mol^{-1}
Enthalpy change of hydration of Cl^- = −378 kJ mol^{-1}
Lattice energy of NH_4Cl = −705 kJ mol^{-1}

(2)

(iv) A student suggested that the enthalpy change of solution of ammonium bromide would be of a similar magnitude to the enthalpy change of solution of ammonium chloride.

Comment on this suggestion in terms of the hydration enthalpies and lattice energies of these two compounds.

(3)

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(c) Write an **ionic** equation to show why aqueous solutions of ammonium chloride are acidic. State symbols are not required.

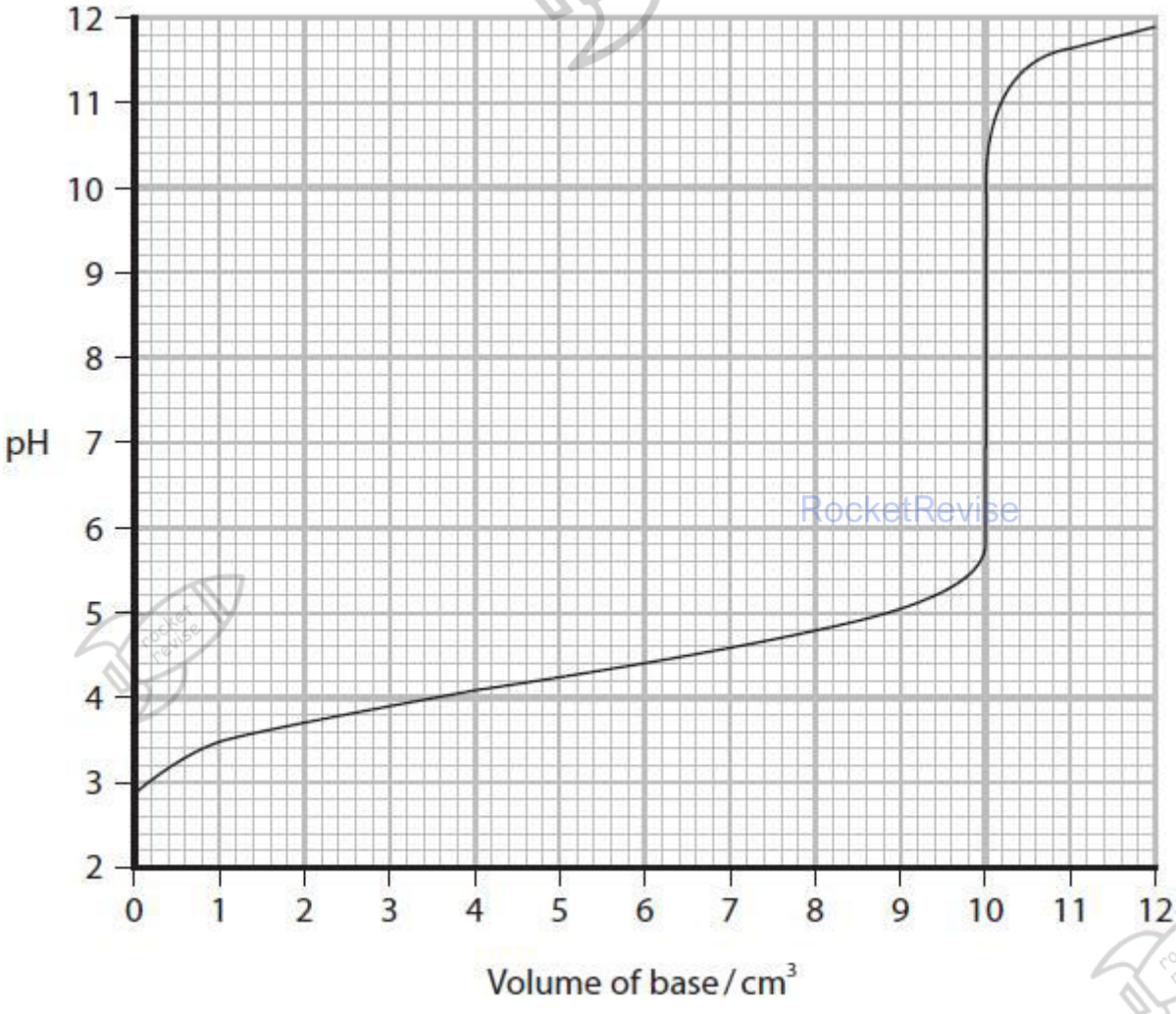
(1)

(Total for question = 17 marks)

Q27.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The titration curve shown is produced when a base is added to an acid.



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(a) Which indicators could be used for this titration? Use your Data Booklet.

(1)

- ☒ **A** bromocresol green, methyl red and phenolphthalein
- ☒ **B** bromothymol blue, phenol red and phenolphthalein
- ☒ **C** methyl red, bromothymol blue and phenol red
- ☒ **D** thymol blue, screened methyl orange and bromophenol blue

(b) Which acid and base could produce this curve?

(1)

- ☒ **A** CH_3COOH and NaOH
- ☒ **B** CH_3COOH and NH_3
- ☒ **C** HCl and NaOH
- ☒ **D** HCl and NH_3

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(Total for question = 2 marks)

Q28.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The solubility of magnesium hydroxide is 1.1×10^{-4} mol dm^{-3} at 25 °C and 4.0×10^{-3} mol dm^{-3} at 100 °C.

[$K_w = 1.0 \times 10^{-14}$ mol^2 dm^{-6} at 25 °C $K_w = 5.5 \times 10^{-14}$ mol^2 dm^{-6} at 100 °C]

(a) What is the pH of a saturated solution of magnesium hydroxide at 25 °C?

(1)

- ☒ **A** 9.7
- ☒ **B** 10.0
- ☒ **C** 10.3
- ☒ **D** 12.0

(b) Saturated solutions of magnesium hydroxide at 25 °C and 100 °C are compared.
How do the concentrations of hydroxide ions and the pH values of these solutions differ?

(1)

	Concentration of hydroxide ions at 100 °C	pH at 100 °C
☒ **A**	higher	lower
☒ **B**	higher	higher
☒ **C**	lower	higher
☒ **D**	lower	lower

(Total for question = 2 marks)

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Q29.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

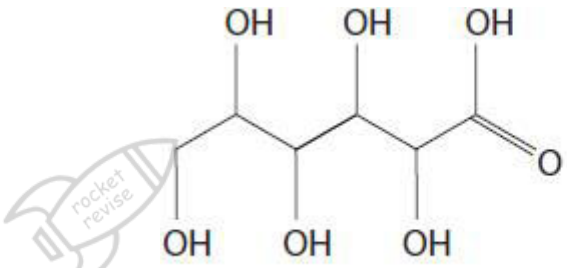
Which is **not** a conjugate acid-base pair?

- ☐ **A** NH_3, NH_2^-
- ☐ **B** NH_4^+, NH_3
- ☐ **C** H_2CO_3, CO_3^{2-}
- ☐ **D** H_2CO_3, HCO_3^-

(Total for question = 1 mark)

Q30.

Gluconic acid is a weak acid which occurs in fruit and honey and is widely used as a food additive.



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gluconic acid

(a) A solution of gluconic acid was prepared by dissolving 4.75 g of the solid in water and making the solution up to 250.0 cm^3 in a volumetric flask.

(i) Write the equation for the acid dissociation constant, K_a, of gluconic acid.
Use RCOOH for gluconic acid.

(1)

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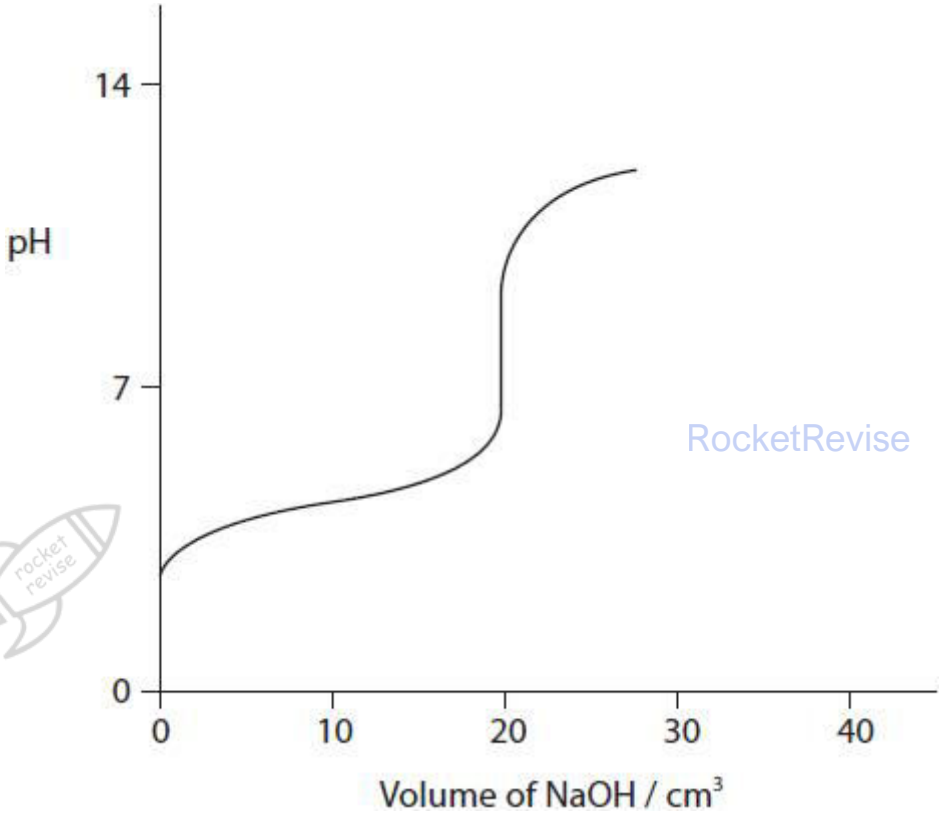
(ii) Calculate the pH of the solution of gluconic acid.
[M_r (gluconic acid) = 196 K_a (gluconic acid) = 1.38×10^{-4} mol dm^{-3}]

(4)

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(b) A solution of 0.105 mol dm^{-3} sodium hydroxide was titrated against 25.0 cm^3 of the gluconic acid solution prepared in (a).

The titration curve is shown. The curve is **not** to scale.



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(i) Explain why phenol red is an appropriate indicator for this titration. Refer to the titration curve and the Data Booklet.

(2)

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(ii) Calculate the pH of the titration mixture after the addition of 35.0 cm^3 of sodium hydroxide solution. The value required **cannot** be obtained from the titration curve in (b).

(5)

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(c) Mixtures of gluconic acid and sodium gluconate are buffers.

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(i) Explain how a solution that contains both gluconic acid and sodium gluconate acts as a buffer. Use RCOOH for the formula of gluconic acid.

(4)

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(ii) The concentration of a solution of gluconic acid is 1.55 mol dm^{-3}.
Calculate the number of moles of sodium gluconate that must be added to 1.00 dm^3 of this solution to give a buffer with pH = 3.71.
Assume that the volume of the solution is unchanged.

(3)

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(Total for question = 19 marks)

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Q31.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the pH of the solution when 2.15 g of barium hydroxide is dissolved in 200 cm^3 of deionised water?

[molar mass of barium hydroxide = 171.3 g mol^{-1} $K_w = 1.00 \times 10^{-14}$ mol^2 dm^{-6}]

- ☐ **A** 10.1
- ☐ **B** 12.1
- ☐ **C** 12.8
- ☐ **D** 13.1

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(Total for question = 1 mark)

Q32.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A 1 mol dm^{-3} solution of ethanoic acid is gradually diluted by the addition of distilled water.

What happens to the degree of dissociation of the acid and the pH of the solution as the distilled water is added?

	Degree of dissociation	pH
☐ A	increases	increases
☐ B	increases	decreases
☐ C	decreases	increases
☐ D	decreases	decreases

(Total for question = 1 mark)

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Q33.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Equal amounts of ethanoic acid and methanoic acid are mixed.

[pK_a values: ethanoic acid = 4.8 methanoic acid = 3.8]

What are the conjugate acid-base pairs in the mixture?

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		acid 1	base 1	acid 2	base 2
☐	A	HCOOH	$HCOO^-$	$CH_3COOH_2^+$	CH_3COOH
☐	B	CH_3COOH	CH_3COO^-	$HCOOH_2^+$	HCOOH
☐	C	HCOOH	CH_3COOH	$CH_3COOH_2^+$	$HCOO^-$
☐	D	CH_3COOH	HCOOH	$HCOOH_2^+$	CH_3COO^-

(Total for question = 1 mark)

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Q34.

This question is about acids.

(a) Calculate the pH of the following acidic solutions. Give your answers to **two** decimal places.

(i) 0.14 mol dm^{-3} solution of hydrochloric acid

(1)

(ii) 0.14 mol dm^{-3} solution of ethanoic acid (K_a = 1.76 × 10^{-5} mol dm^{-3})

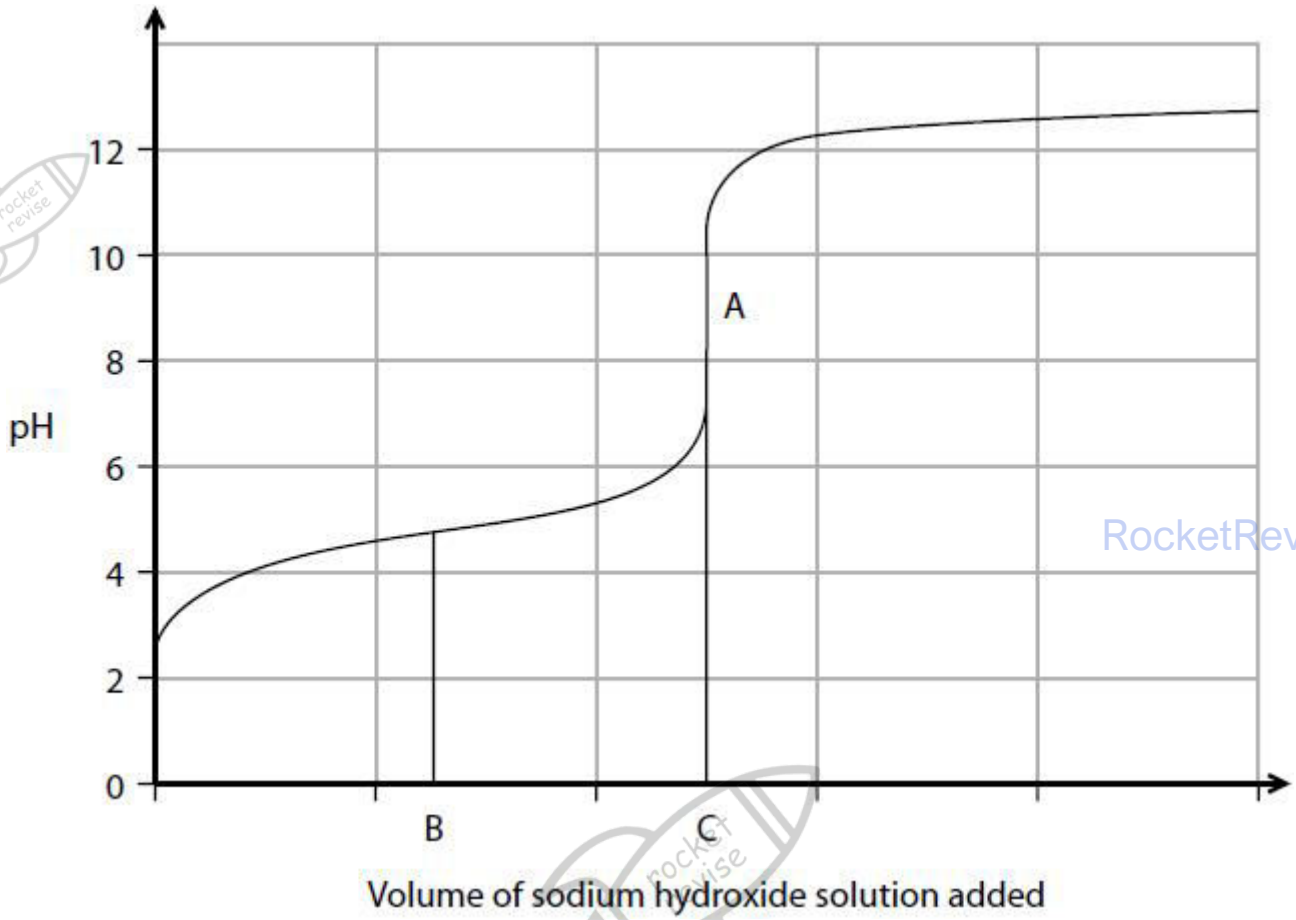
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(b) The graph shows the titration curve for a weak acid with a strong base. The equivalence point is A and the volume of alkali added at the equivalence point is C. Volume B is half of volume C.

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(i) Use the graph to determine the dissociation constant, K_a, of the weak acid.

(3)

(ii) Explain the rapid rise in pH from 2.8 to 4 at the start of the titration.

(2)

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(c) In another experiment, 10 cm^3 of 1.0 mol dm^{-3} sodium hydroxide was added to 30 cm^3 1.0 mol dm^{-3} propanoic acid (K_a = 1.3 × 10^{-5} mol dm^{-3}).

(i) Calculate the pH of the resulting solution.

(3)

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(ii) State one assumption you have made in this calculation.

(1)

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(Total for question = 13 marks)

Q35.

Which indicator should be used to determine the end point in a titration of a strong acid with a weak base?

(1)

- ☒ **A** universal indicator
- ☒ **B** methyl orange
- ☒ **C** phenolphthalein
- ☒ **D** litmus

(Total for question = 1 mark)

Q36.

The equation for the dissociation of water is:

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$$H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$$

The ionic product of water, K_w, varies with temperature.

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Temperature/ °C	K_w / mol^{2}dm^{-6}
25	1.01×10^{-14}
30	1.47×10^{-14}
50	5.48×10^{-14}
100	7.16×10^{-14}

What is the pH of pure water at 60 °C?

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(1)

- ☒ **A** approximately 6.5
- ☒ **B** exactly 7
- ☒ **C** approximately 7.4
- ☒ **D** greater than 7.4

(Total for question = 1 mark)

Q37.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Ethanoic acid, CH_3COOH, can react with a hydrogensulfite ion, HSO_3^- .

Which is a correct acid conjugate-base pair for this reaction?

[K_a for ethanoic acid = 1.7×10^{-5} mol dm^{-3}

K_a for hydrogensulfite ion = 6.2×10^{-8} mol dm^{-3}]

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	Acid	Conjugate-base
☒ **A**	HSO_3^-	SO_3^{2-}
☒ **B**	HSO_3^-	H_2SO_3
☒ **C**	CH_3COOH	$CH_3CO_2^-$
☒ **D**	CH_3COOH	$CH_3COOH_2^+$

(Total for question = 1 mark)

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Q38.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box () and then mark your new answer with a cross (☒).

Which indicator can be used to show the end-point of a titration between hydrochloric acid, HCl(aq), and ammonia solution, $NH_3(aq)$?

- ☒ **A** universal indicator
- ☒ **B** phenolphthalein
- ☒ **C** phenol red
- ☒ **D** methyl orange

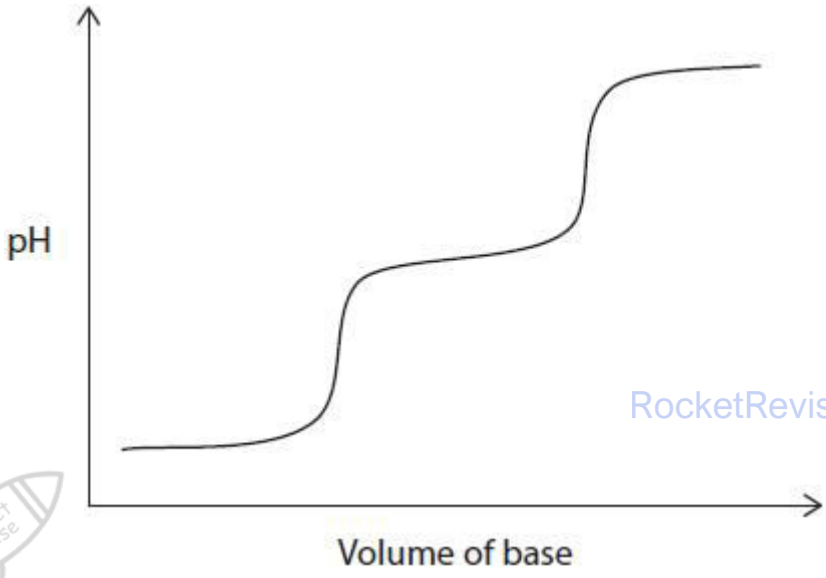
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(Total for question = 1 mark)

Q39.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

A titration curve for the titration of an acid with a strong base is shown.



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What could be the formula of the acid?

- ☒ **A** HX
- ☒ **B** HX_2
- ☒ **C** H_2X
- ☒ **D** H_3X

(Total for question = 1 mark)

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Q40.

This question is about acids and bases.

(a) Some K_a and pK_a values for several weak acids are shown.

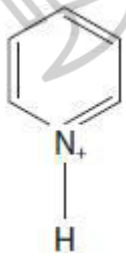
Weak acid	K_a / mol dm^{-3}	pK_a
$C_5H_5NH^+$		5.25
CH_3CH_2COOH	1.3×10^{-5}	4.88
$HCOOH$	1.8×10^{-4}	3.75
$CH_2ClCOOH$	1.4×10^{-3}	2.85
$CHCl_2COOH$	4.5×10^{-2}	

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(i) Complete the table.

(2)

(ii) The structure of $C_5H_5NH^+$ is shown.



Write the K_a expression for this weak acid.

(1)

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(iii) When methanoic acid is added to propanoic acid, an equilibrium is set up containing two acid-base pairs.

Complete the equilibrium, labelling the acid-base pairs as **A1**, **B1** and **A2**, **B2**.

(2)

$$CH_3CH_2COOH + HCOOH \rightleftharpoons \text{.....} + \text{.....}$$

....

*(iv) A student measured the pH of separate 0.050 mol dm^{-3} solutions of $CH_2ClCOOH$ and of

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$CHCl_2COOH$, using a pH meter.

The student also calculated the pH of these solutions, making two assumptions:

- $[H^+] = [A^-]$
- $[HA]_{equilibrium} = [HA]_{initial}$

The measured and calculated pH values are shown.

	0.050 mol dm^{-3} $CH_2ClCOOH$	0.050 mol dm^{-3} $CHCl_2COOH$
Measured pH value	2.11	1.52
Calculated pH value	2.08	1.32

Discuss the differences between the student's measured and calculated pH values.

(6)

In your answer, you should

- show how the student calculated their pH values

[Ka($CH_2ClCOOH$) = 1.4×10^{-3} mol dm^{-3};
Ka($CHCl_2COOH$) = 4.5×10^{-2} mol dm^{-3}]

- explain, with reference to the assumptions made, why there is a difference between the calculated and measured pH values
- suggest why the measured pH values are higher than the calculated pH values.

(b) 4-Hydroxybenzoic acid, HOC_6H_4COOH, is a diprotic acid, which dissociates in two stages. Each stage has a different pK_a value.

Stage 1 $HOC_6H_4COOH \rightleftharpoons H^+ + HOC_6H_4COO^-$ $pK_{a1} = 4.5$

Stage 2 $HOC_6H_4COO^- \rightleftharpoons H^+ + {}^-OC_6H_4COO^-$ $pK_{a2} = 9.3$

In a titration, 75.0 cm^3 of 0.025 mol dm^{-3} NaOH was added to 25.0 cm^3 of 0.025 mol dm^{-3} HOC_6H_4COOH.

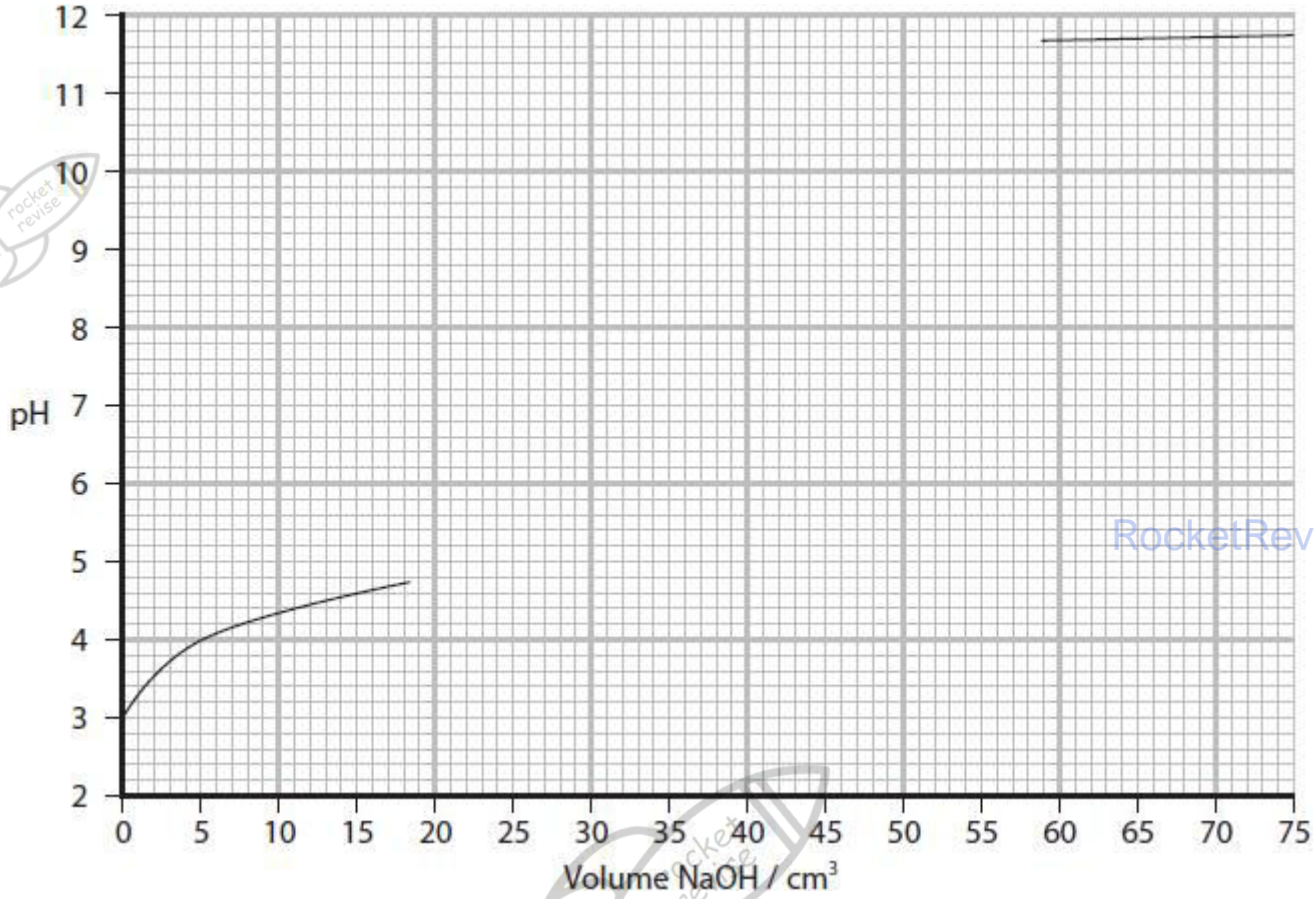
Complete the titration curve.

(3)

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(c) A student was given 50.0 cm^3 of a solution of sodium hydroxide.

The pH of this solution was 12.43.
The student was asked to adjust the pH to 12.00, by dilution with deionised water.
The student did **not** have access to a pH meter.
Calculate the volume of deionised water, **in cm^3**, the student should add to the original solution.

(5)

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(Total for question = 19 marks)

Q41.

This question is about acids containing sulfur.

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(a) Sulfuric acid reacts with sodium hydroxide solution as shown.

$$H_2SO_4(aq) + 2NaOH(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$$

Calculate the pH of the solution formed when 20.0 cm^3 of sulfuric acid of concentration = 0.0720 mol dm^{-3} is mixed with 80.0 cm^3 of sodium hydroxide solution of concentration = 0.240 mol dm^{-3} .
Give your answer to one decimal place.

$$K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$$

(6)

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(b) Sulfurous acid, $H_2SO_3(aq)$, is a weak acid and dissociates as shown.

$$H_2SO_3(aq) \rightleftharpoons HSO_3^-(aq) + H^+(aq)$$

It forms a buffer solution when mixed with sodium hydrogensulfite, $NaHSO_3$.

(i) State what is meant by the term weak in this context.

(1)

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(ii) State what is meant by the term buffer solution.

(2)

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(iii) Calculate the mass of sodium hydrogensulfite that should be added to 50.0 cm^3 of 0.0480 mol dm^{-3} sulfurous acid solution to form a buffer solution of pH = 2.18

[K_a (H_2SO_3) = 1.54×10^{-2} mol dm^{-3} Relative formula mass of $NaHSO_3$ = 104.1]

(5)

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(Total for question = 14 marks)

Q42.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

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Phenolphthalein is a very weak acid and it forms a colourless solution under acidic conditions.

Sodium hydroxide is titrated against hydrochloric acid using phenolphthalein indicator. At the end-point, the solution is pink because phenolphthalein

- ☐ **A** accepts hydroxide ions and becomes a cation
- ☐ **B** accepts hydroxide ions and becomes an anion
- ☐ **C** donates hydrogen ions and becomes a cation
- ☐ **D** donates hydrogen ions and becomes an anion

(Total for question = 1 mark)

Q43.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Equimolar solutions of four acids are prepared. Which solution has the **lowest** pH?

Use electronegativity values from the Data Booklet.

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- ☐ **A** CH_3COOH
- ☐ **B** $CH_2ClCOOH$
- ☐ **C** $CH_2BrCOOH$
- ☐ **D** CH_2ICOOH

(Total for question = 1 mark)

Q44.

Answer the question with a cross in the box you think is correct ☒. If you change your mind

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about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Some equations for acid-base equilibria are shown.

$$H_3PO_4 + H_2O \rightleftharpoons H_2PO_4^- + H_3O^+$$

$$H_2PO_4^- + H_2O \rightleftharpoons HPO_4^{2-} + H_3O^+$$

$$HPO_4^{2-} + H_2O \rightleftharpoons PO_4^{3-} + H_3O^+$$

What is the conjugate acid of HPO_4^{2-}?

- ☒ **A** H_3PO_4
- ☒ **B** H_3O^+
- ☒ **C** $H_2PO_4^-$
- ☒ **D** PO_4^{3-}

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(Total for question = 1 mark)

Q45.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

The dissociation of water is an endothermic reaction.

$$H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq) \quad \Delta H \text{ is positive}$$

[$K_w = 1.0 \times 10^{-14}\,mol^2\,dm^{-6}$ at 298 K $K_w = 2.92 \times 10^{-14}\,mol^2\,dm^{-6}$ at 313 K]

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What is the best **estimate** of the pH of water at 313 K?

- ☒ **A** 7.8
- ☒ **B** 7.0
- ☒ **C** 6.8
- ☒ **D** 5.8

(Total for question = 1 mark)

Q46.

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Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

The accumulation of carbon dioxide during exercise reduces the pH in cells and in blood.

Which of these reactions is **most** important in buffering the decrease in pH that occurs?

- ☐ **A** $H_2CO_3 \rightarrow H^+ + HCO_3^-$
- ☐ **B** $HCO_3^- + H^+ \rightarrow H_2CO_3$
- ☐ **C** $HCO_3^- \rightarrow H^+ + CO_3^{2-}$
- ☐ **D** $CO_3^{2-} + H^+ \rightarrow HCO_3^-$

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(Total for question = 1 mark)

Q47.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

At 50 °C, the ionic product of water, K_w, is 5.5×10^{-14} mol^2 dm^{-6}.

At this temperature, water is

- ☐ **A** neutral with a pH of 7.0
- ☐ **B** neutral with a pH of 6.6
- ☐ **C** acidic with a pH of 6.6
- ☐ **D** alkaline with a pH of 7.4

(Total for question = 1 mark)

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Q48.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

A buffer solution contains CH_3COOH and CH_3COONa in the ratio, 1 : 4 .

What is the pH of this buffer solution?

[$K_a(CH_3COOH) = 1.7 \times 10^{-5}$ mol dm^{-3}]

- ☐ **A** 2.4
- ☐ **B** 4.2

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- ☒ **C** 4.8
- ☒ **D** 5.4

(Total for question = 1 mark)

Q49.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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What is the pH of a 0.200 mol dm^{-3} solution of strontium hydroxide, $Sr(OH)_2$?

$$Kw = 1.00 \times 10^{-14}\ mol^2\ dm^{-6}$$

- ☒ **A** 14.0
- ☒ **B** 13.6
- ☒ **C** 13.3
- ☒ **D** 12.6

(Total for question = 1 mark)

Q50.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the pH of a 0.100 mol dm^{-3} solution of sodium hydroxide at 283 K?

[pK_w = 14.53 at 283 K]

- ☒ **A** 13.00
- ☒ **B** 13.53
- ☒ **C** 14.43
- ☒ **D** 14.53

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(Total for question = 1 mark)

Q51.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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A strong monoprotic acid, of concentration 0.0100 mol dm^{-3}, has a pH = 2.0.

What is the **decrease in pH** if the concentration is increased to 0.0110 mol dm^{-3}?

- ☒ **A** 0.001
- ☒ **B** 0.04
- ☒ **C** 1.04
- ☒ **D** 1.96

(Total for question = 1 mark)

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Q52.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The pH of a dilute solution of a weak carboxylic acid can be calculated from its K_a value.

Which is **not** a relevant assumption in the calculation of pH?

- ☒ **A** the ionisation of water is insignificant
- ☒ **B** the initial and equilibrium concentrations of the acid are the same
- ☒ **C** the concentrations of the protons and carboxylate ions are the same
- ☒ **D** the dissociation of the carboxylic acid is almost complete

(Total for question = 1 mark)

Q53.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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This is a question about buffers.

(a) The Henderson–Hasselbalch equation can be used to determine the pH of a buffer solution. This equation is shown.

$$pH = pK_a + \log\frac{[salt]}{[acid]}$$

What is the pH of a buffer solution containing 0.100 mol dm^{-3} ethanoic acid and 0.200 mol dm^{-3} sodium ethanoate?
[K_a(ethanoic acid) = 1.7×10^{-5} mol dm^{-3} at 298 K]

(1)

- ☒ **A** 4.47
- ☒ **B** 4.77
- ☒ **C** 5.07

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☒ **D** 5.46

(b) What is the **best** reason for the use of buffers in foods?

(1)

☒ **A** to improve taste
☒ **B** to help digestion of the food
☒ **C** to prevent deterioration by fungal activity
☒ **D** to enhance the colour and appearance of the food

(c) The carbonic acid/hydrogencarbonate ion buffer system in the blood involves the equilibria shown.

$$CO_2(g) + H_2O(l) \rightleftharpoons H_2CO_3(aq) \rightleftharpoons H^+(aq) + HCO_3^-(aq)$$

How is the pH of blood affected by increased exercise?

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(1)

☒ **A** there is no effect because the equilibria are not affected by activity
☒ **B** the pH will decrease due to the equilibria shifting to the left
☒ **C** the pH will decrease due to the equilibria shifting to the right
☒ **D** the pH is constant because it is a buffer system

(Total for question = 3 marks)

Q54.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

When ammonia dissolves in water, an equilibrium results. The equation for this equilibrium is shown.

$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$$

Which of these species act as a Brønsted–Lowry acid in this equilibrium?

☐ **A** H_2O only
☐ **B** NH_3 and OH^-
☐ **C** NH_4^+ only
☐ **D** NH_4^+ and H_2O

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(Total for question = 1 mark)

Q55.

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This question is about weak acids.

The values for the acid dissociation constant of ethanoic acid and chloroethanoic acid are shown.

Acid	K_a / mol dm^{-3}
ethanoic acid	1.70×10^{-5}
chloroethanoic acid	1.30×10^{-3}

(a) (i) Write the expression for K_a for ethanoic acid, CH_3COOH.

(1)

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(ii) Calculate the pH of a 0.25 mol dm^{-3} solution of ethanoic acid.

(2)

(iii) State **one** assumption you have made when calculating the pH in (a)(ii).

(1)

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(b) (i) Suggest a reason why chloroethanoic acid, $CH_2ClCOOH$, is a stronger acid than ethanoic acid.

(1)

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(ii) When ethanoic acid and chloroethanoic acid are added together an equilibrium is set up containing two acid-base pairs.

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Complete the equilibrium equation, labelling the conjugate acid-base pairs as A1, B1 and A2, B2.

(2)

$CH_3COOH + CH_2ClCOOH \rightleftharpoons$

(c) A buffer was made by mixing 8.20 g of sodium ethanoate, CH_3COONa, with 250 cm^3 of 0.700 mol dm^{-3} solution of ethanoic acid, CH_3COOH.

Calculate the pH of the buffer.

(4)

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(d) A type of bubble bath contains a weak acid and the indicator bromocresol green.

The bubble bath is initially yellow but when diluted with bath water, its colour changes from yellow to green to blue.
Use the information on page 9 of the Data Booklet to deduce why the bubble bath changes colour.

(3)

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(Total for question = 14 marks)

Q56.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A sodium hydroxide solution with pH = 12 is diluted by a factor of 100.

What is the pH of the diluted solution?

- ☒ **A** 10
- ☒ **B** 11
- ☒ **C** 13
- ☒ **D** 14

(Total for question = 1 mark)

Q57.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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What is the pH of a 1.25 mol dm^{-3} solution of hydrochloric acid?

- ☒ **A** −1.25
- ☒ **B** −0.097
- ☒ **C** 0.097
- ☒ **D** 1.25

(Total for question = 1 mark)

Q58.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

At 25°C, the pH of pure water is 7.0 and at 75°C the pH is 6.4.

Why is the pH of pure water lower at 75°C than at 25°C?

- ☒ **A** there are more hydrogen ions than hydroxide ions when the water is at 75°C
- ☒ **B** the dissociation of water is endothermic so the concentration of hydrogen ions is lower at 75°C
- ☒ **C** the dissociation of water is endothermic so the concentration of hydrogen ions is higher at 75°C
- ☒ **D** the dissociation of water is exothermic so the concentration of hydrogen ions is higher at 75°C

(Total for question = 1 mark)

Q59.

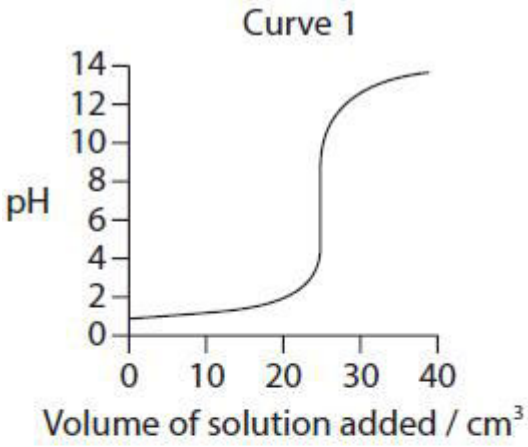
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒. and then mark your new answer with a cross ☒.

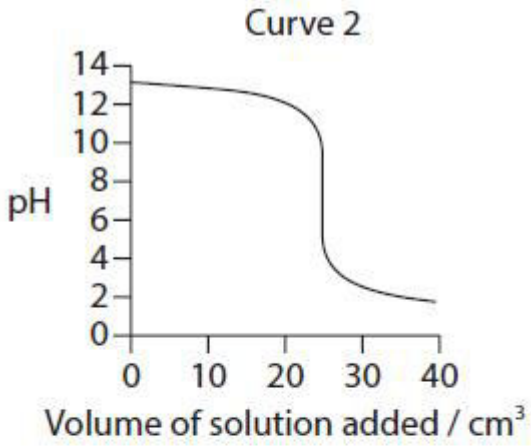
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This question is about the titration curves obtained by reacting different acids and alkalis of the same concentration.

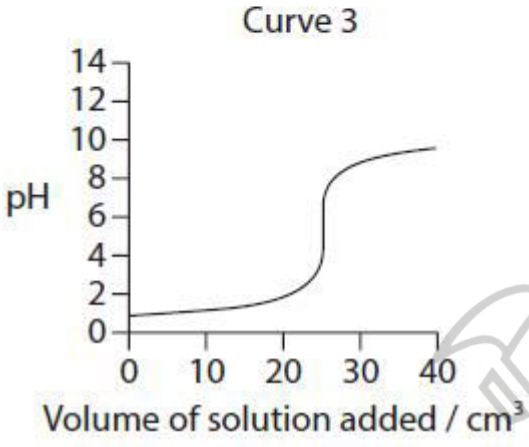
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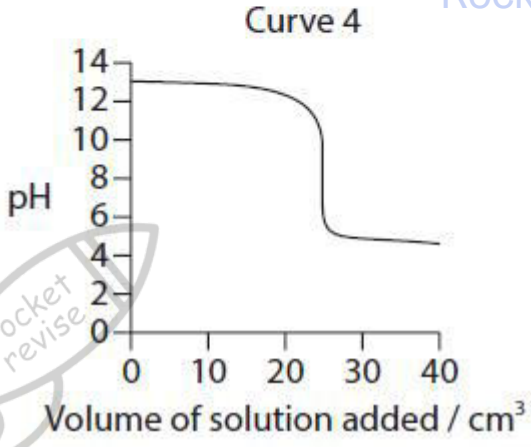
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(a) Which curve is produced by adding ammonia to 25cm^3 of hydrochloric acid?

(1)

- ☒ **A** Curve 1
- ☒ **B** Curve 2
- ☒ **C** Curve 3
- ☒ **D** Curve 4

(b) Which of these indicators would be **most** suitable to determine the end-point for the titration that gives Curve 4?

(1)

	Indicator	pH range
☒ **A**	congo red	3.0–5.0
☒ **B**	bromocresol purple	5.2–6.8
☒ **C**	thymol blue (base)	8.0–9.6
☒ **D**	alizarin yellow R	10.1–13.0

(Total for question = 2 marks)

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